

Challenge 4

Crop-Specific vs. Crop-Independent Farmer Questions

Goal

Understand the balance between crop-specific questions and general agricultural concerns.

This analysis explores how smallholder farmers frame their information needs on the WeFarm platform, focusing on whether questions are tied to specific crops or animals, or reflect broader, crop-independent agricultural concerns.

Background: Company and Domain Context

Producers Direct is a UK-based non-profit organization operating in the agricultural development and digital services sector. The organization supports over one million smallholder farmers across East Africa and Latin America through digital tools for market access, peer-to-peer knowledge sharing, and data-informed decision making.

Producers Direct acquired the **WeFarm dataset** after WeFarm ceased operations in late 2022. WeFarm operated for approximately seven years as an SMS-based knowledge-sharing platform that allowed farmers to ask agricultural questions and receive peer responses. During its operation, the platform accumulated millions of questions and responses across multiple countries and languages.

From the perspective of a **volunteer data analyst working with Producers Direct**, this dataset represents **farmer demand for agricultural information**, expressed through millions of farmer-initiated questions rather than commercial transactions. Core analytical signals include question volume, topic distribution, repetition, response specificity, and variation across time and geography. Analyzing the balance between crop-specific and general questions helps inform how agricultural knowledge services should be structured, prioritized, and scaled to meet recurring farmer information needs.

Key Analytical Areas

Insights are organized across four categories that directly map to the Challenge 4 guiding questions:

- **Category 1:** Question Type Distribution
- **Category 2:** Topic Concentration and Cross-Cutting Themes
- **Category 3:** Temporal and Geographic Variation
- **Category 4:** Repetition and Response Specificity

SQL queries used for data inspection and preparation:

[SQL Data Cleaning and Preparation – link]

Targeted SQL queries used to answer Challenge 4 questions:

[Challenge 4 Analysis Queries – link]

Data Structure and Initial Checks

The analytical dataset is derived from the original WeFarm relational data, which contains question and response records collected during the platform's operation. Several derived views and tables were created to support this analysis.

- **farmer_questions:** Original response-level table containing question content, response content, language, country, topic labels, and timestamps.
- **question_level (derived view):** One row per unique question identifier, created to analyze question volume without inflating counts due to multiple responses.
- **challenge4_labeled (derived table):** Question-level table assigning each question a question type (crop-specific, livestock-specific, general, or unknown).
- **response_level (derived table):** Response-level table joining WeFarm responses to question metadata, used for response behavior and specificity analysis.

The analysis covers approximately **5.8 million unique questions** and **17 million responses**. Aggregation to the question level was required to avoid overstating question volume for questions that received many responses.

Executive Summary

Analysis of more than **4.1 million farmer questions** shows that information demand is highly concentrated, with the **top 15 crop and animal topics accounting for 73.4 percent of all questions**, led by **maize (535,569 questions)**, **chicken (430,894)**, and **cattle (425,026)**. Although **general, crop-independent questions make up a smaller share of total volume (about 9–12 percent per month)**, they consistently focus on cross-cutting needs such as **planting (29.8 percent of general questions)**, along with disease, pests, and soil management, highlighting persistent baseline information needs across crops. Over time and across countries, **crop-specific questions dominate overall volume**, but all question types rise and fall together seasonally, indicating that **general questions complement rather than replace crop-specific demand**.

Insights Deep Dive

Category 1: Question Type Distribution

Guiding Questions Answered

What proportion of questions are crop-specific vs. general?

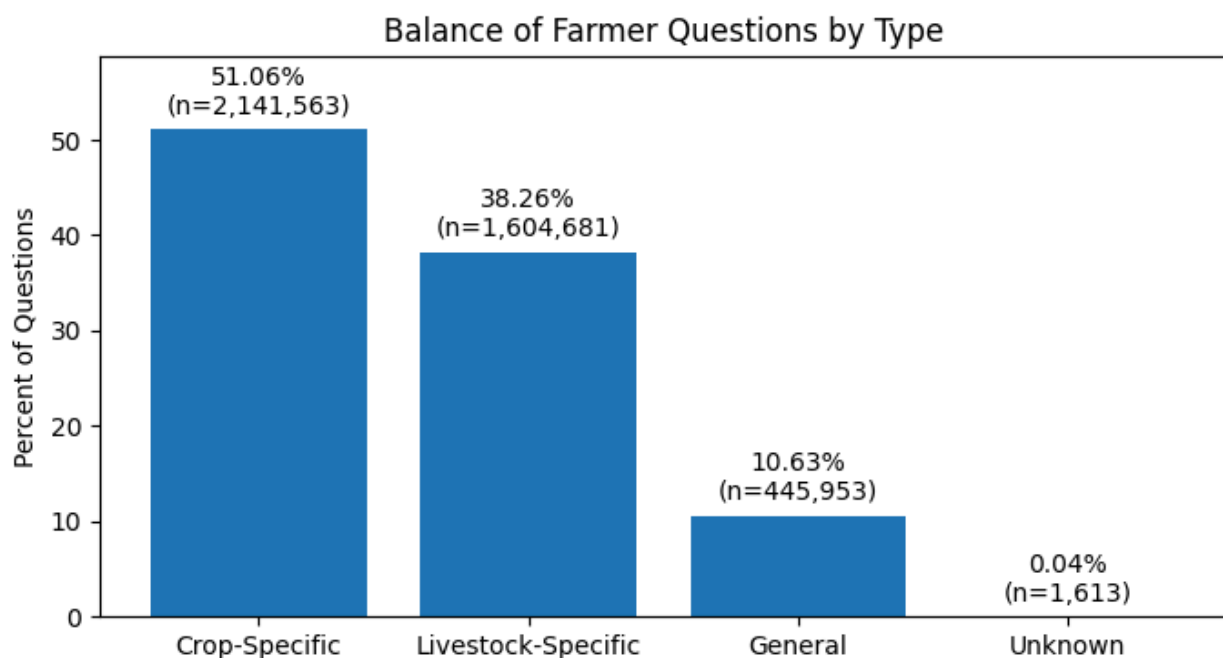
Deliverables Used

- Question-type distribution bar chart
- Question-level aggregation table

Key Insights

- **Farmer questions are overwhelmingly driven by specific, hands-on needs**, as just over **half of all questions (51.06%, about 2.1 million)** focus on a single crop, showing that farmers primarily come to the platform looking for practical guidance they can apply directly to what they are growing right now.
- **Livestock questions form a strong second pillar of demand**, making up **38.26% (about 1.6 million questions)**, which indicates that animal production is a consistently important and recurring concern rather than a niche topic, nearly matching crop-related demand in scale.

- **General agricultural questions play a smaller, supporting role, at 10.63% (about 446,000 questions),** suggesting that farmers rely on broad guidance mainly as background knowledge while prioritizing crop- and animal-specific advice for day-to-day decisions.
- **The combined dominance of crop- and livestock-specific questions (over 89%) shows a clear pattern,** where farmers return repeatedly with highly targeted questions, reflecting long-term reliance on detailed, situation-specific information rather than one-time, high-level explanations.
- **The extremely small share of unknown questions (0.04%) indicates stability in how farmers' needs are categorized,** reinforcing confidence that the observed pattern is real and persistent, not an artifact of missing or unclear data.



Category 2: Topic Concentration and Cross-Cutting Themes

Guiding Questions Answered

What topics cut across all crops?

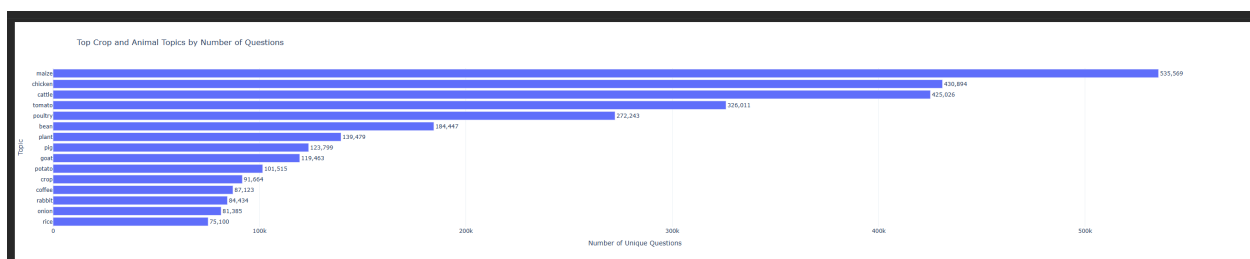
Are questions concentrated around specific crops or animals?

Deliverables Used

- Topic frequency tables
- Cross-cutting theme classification table

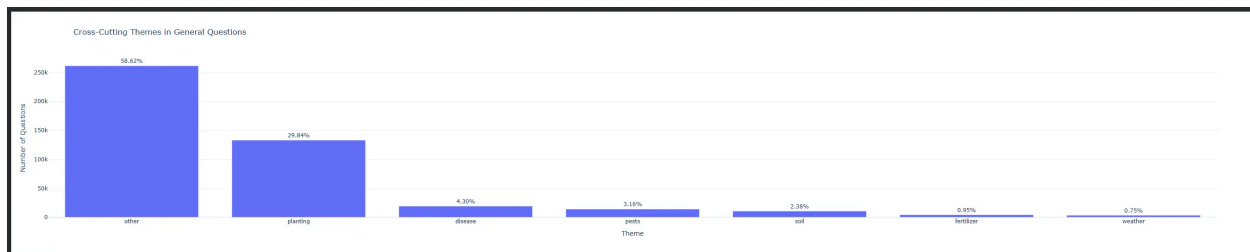
Key Insights

- **A few staple crops and animals dominate farmer demand**, with maize alone generating **535,569 questions**, and chicken (**430,894**) and cattle (**425,026**) close behind. These three topics together account for **over 1.39 million questions**, showing that farmers repeatedly seek guidance on the same core production areas rather than spreading questions evenly across all crops and animals.
- **The top five topics (maize, chicken, cattle, tomato, and poultry) each exceed 270,000 questions**, while most remaining topics fall below **200,000 questions**, revealing a sharp drop-off after the highest-priority crops and animals. This steep decline tells a story of long-term concentration, where farmer engagement is driven by a small, consistent set of high-impact topics.
- **Livestock topics are nearly as prominent as crops at the very top of the distribution**, with chicken and cattle each approaching maize in volume. This shows that animal production has been a persistent and recurring source of farmer questions, not a secondary or occasional concern.



- **Planting is the most prominent cross-cutting theme**, accounting for **29.84% of all general questions**, indicating that farmers regularly ask about timing and setup issues that recur across seasons and apply regardless of crop type.

- **More than half of the general questions fall into a broad “other” category (58.62%),** suggesting that many farmers’ concerns are multi-dimensional and cannot be easily reduced to a single technical issue. This pattern reflects ongoing planning and context-specific challenges rather than isolated, one-off problems.
- **Problem-focused themes such as disease (4.30%), pests (3.16%), and soil (2.38%) appear consistently but at lower levels,** showing that these issues affect all crops but tend to arise as background concerns rather than the main driver of general inquiries.
- **Weather (0.75%) and fertilizer (0.95%) questions make up less than 1% each,** implying that although these factors influence all farming, farmers usually consider them as part of broader planning questions rather than asking about them in isolation.



Category 3: Temporal and Geographic Variation

Guiding Questions Answered

How do these question types vary by season or location?

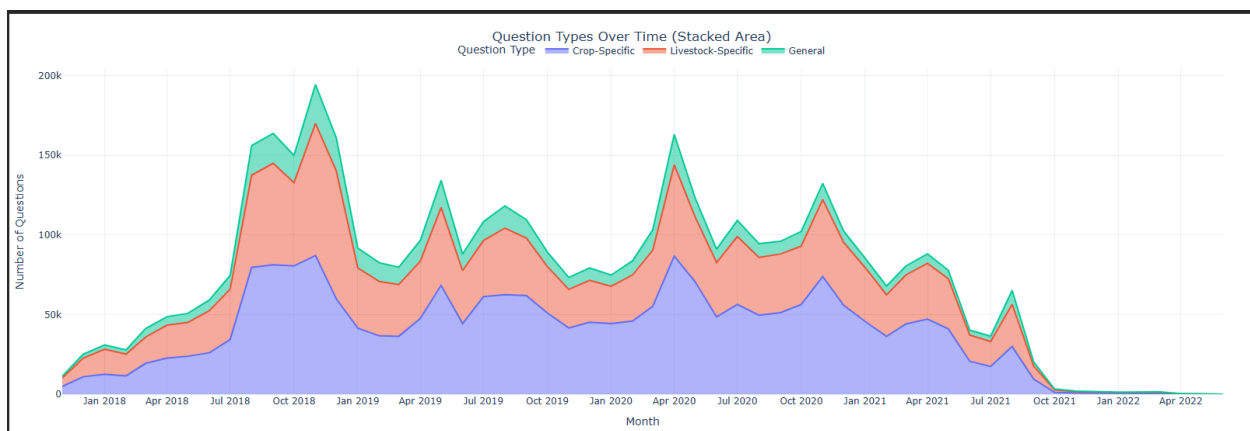
Deliverables Used

- Stacked area chart of question types over time
- Monthly time-series line charts
- Country-level percentage bar charts

How do these questions vary by season?

Key Insights

- **All three question types rise and fall together over time**, with total monthly questions peaking at **roughly 180,000–200,000 questions** during high-activity periods and dropping sharply during low-activity periods. This shared movement shows that farmer engagement is strongly seasonal rather than driven by one specific question type.
- **Crop-specific questions consistently make up the largest share each month**, often contributing **over half of total monthly questions**, especially during peak seasons. This pattern suggests that when farming activity increases, farmers primarily ask questions tied to the crops they are actively growing.
- **Livestock-specific questions follow the same seasonal waves**, adding **tens of thousands of questions per month** during peaks, which indicates that livestock concerns intensify at the same times as crop-related activity rather than operating on a separate seasonal cycle.
- **General (crop-independent) questions remain a small but steady layer**, typically contributing **under 15,000–20,000 questions per month**, even during peak periods. This stability suggests that general planning and management questions persist year-round but do not drive seasonal spikes.



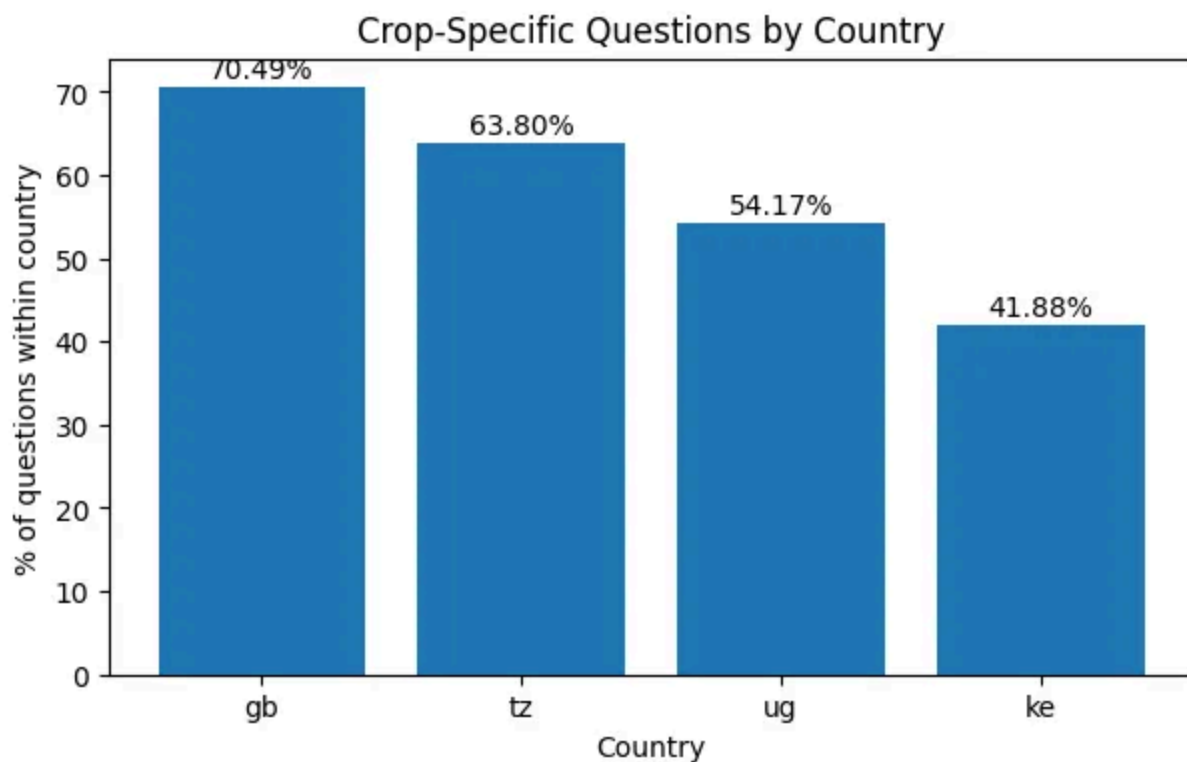
How do question types vary by location?

Crop-specific questions by country

- **Ghana shows the strongest focus on crop-specific questions**, with **70.49%** of all questions tied to specific crops. This suggests that farmers in Ghana

most often seek very targeted, crop-by-crop guidance rather than broad advice.

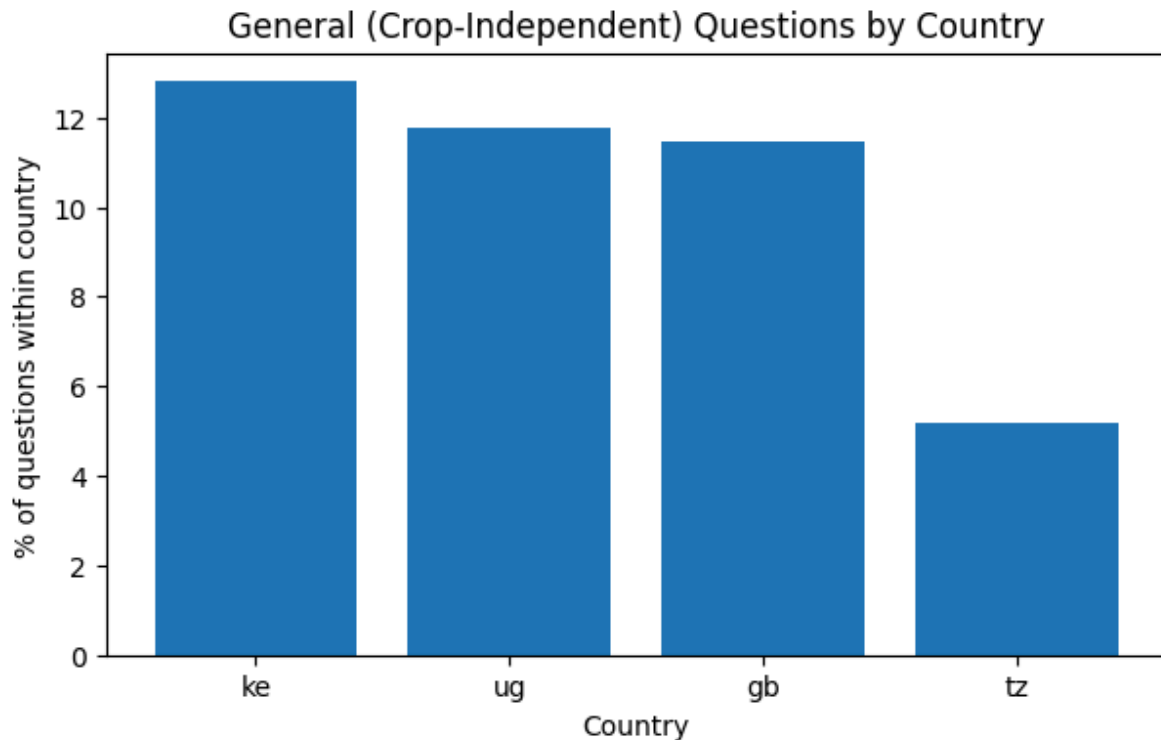
- **Tanzania also has a high share of crop-specific questions**, at **63.80%**, indicating a similar pattern where farmers prioritize detailed guidance for particular crops they are actively managing.
- **Uganda sits in the middle**, with **54.17%** of questions being crop-specific, showing a balance between targeted crop questions and other types of agricultural concerns.
- **Kenya has the lowest share of crop-specific questions**, at **41.88%**, which tells a different story where farmers are less focused on single crops and more likely to ask questions that go beyond one specific crop.



General (crop-independent) questions by country

- **Kenya has the highest share of general questions**, at **12.8%**, suggesting that farmers there more frequently ask broad questions that apply across multiple crops, such as planning, management, or general farming practices.

- **Uganda and Ghana show similar levels of general questions**, at **11.9%** and **11.4%**, indicating a moderate reliance on cross-cutting agricultural guidance alongside crop-specific needs.
- **Tanzania stands out with the lowest share of general questions**, at just **5.2%**, reinforcing the earlier pattern that farmers there tend to focus much more heavily on crop-specific issues rather than broad, cross-cutting topics.



Category 4: Repetition and Response Specificity

Guiding Questions Answered

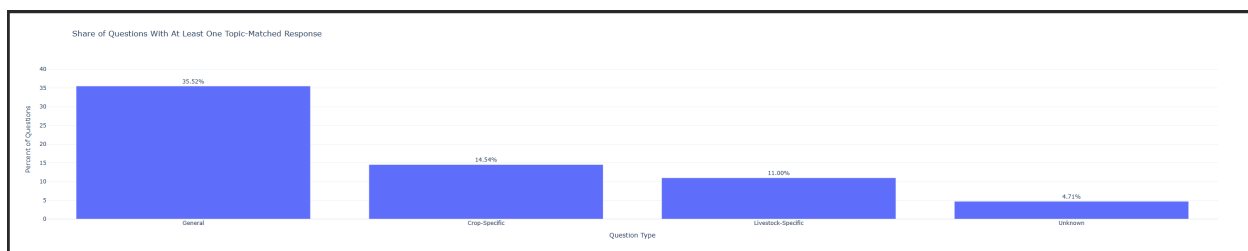
Are certain questions repeated?

How do farmers respond to more general vs. more specific questions?

Key Insights

Are certain questions repeated?

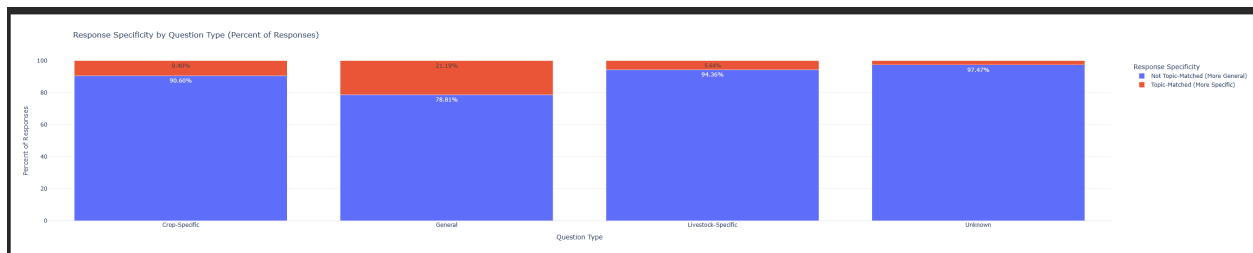
- **General questions are the most likely to receive repeated or multiple related responses**, with **35.52%** of general questions having **at least one topic-matched response**. This higher share suggests that general questions tend to resurface across many farmers and contexts, leading to repeated engagement around similar broad concerns.
- **Crop-specific questions show lower repetition**, with only **14.54%** having at least one topic-matched response. This indicates that while crop-specific questions are very common overall, they are more fragmented across crops and situations rather than being repeated in the same way.
- **Livestock-specific questions show even less repetition**, at **11.00%**, suggesting that animal-related questions are often more situational or context-specific, leading to fewer repeated, closely matched responses.
- **Unknown questions show minimal repetition (4.71%)**, reinforcing that repetition is not driven by noise or misclassification, but by the nature of the question type itself.



How do farmers respond to more general vs. more specific questions?

- **General questions receive the most specific responses**, with **21.19%** of responses being topic-matched. This suggests that when farmers ask broad questions, responses often add specificity by anchoring advice to particular topics or examples.
- **Crop-specific questions receive fewer highly specific responses**, with only **9.40%** of responses being topic-matched, meaning that **over 90%** of responses remain relatively general. This indicates that even when farmers ask about a specific crop, responses often stay broad or high-level.

- **Livestock-specific questions show the lowest response specificity**, with just **5.64%** of responses being topic-matched. This implies that livestock advice is especially likely to be delivered in general terms rather than tightly tailored to the specific animal mentioned.
- **Across all question types, the majority of responses are not topic-matched**, ranging from **78.81% to 97.47%**, showing a consistent pattern where responses tend to emphasize general guidance over narrow, highly specific instructions.



Recommendations

- **Put most effort into crop and livestock help, because that is what farmers ask about most.** Crop-specific questions are **51.06% (2,141,563)** and livestock-specific questions are **38.26% (1,604,681)**, while general questions are only **10.63% (445,953)**.
- **Start with the biggest topics first, because a few crops and animals account for a huge share of questions.** Farmers ask the most about **maize (535,569)**, **chicken (430,894)**, **cattle (425,026)**, **tomato (326,011)**, and **poultry (272,243)**, so building strong guides for these topics will help the most people.
- **Create reusable planting guidance, because planting is the most common general need.** Planting makes up **29.84%** of general questions, which shows many farmers repeatedly need help with when and how to plant across different crops.
- **Make it easier to ask and find answers for broad questions, because most general questions are "mixed" and hard to categorize.** The "other" bucket is **58.62%** of general questions, which suggests many farmers ask multi-part

questions that would benefit from clearer prompts or better navigation to existing answers.

- **Prepare for seasonal spikes, because question volume rises and falls in predictable waves.** The trend chart shows monthly volume peaking around **180,000 to 200,000 questions**, so publishing key guidance before peak periods and planning support ahead of time will help farmers when demand is highest.
 - **Adjust the approach by country, because farmers ask different types of questions in different places.** Crop-specific questions range from **70.49% in Ghana** to **41.88% in Kenya**, and general questions range from **12.8% in Kenya** to **5.2% in Tanzania**, so each country may need a different mix of crop-specific guides and general farming support.
 - **Make answers more specific for crop and livestock questions, because most responses are still too general.** Only **9.40%** of responses to crop-specific questions and **5.64%** of responses to livestock-specific questions are topic-matched, meaning most answers do not directly match the crop or animal mentioned.
 - **Improve access to answers for common general questions, because general questions show the strongest repeated patterns.** General questions have the highest share with at least one topic-matched response at **35.52%**, compared with **14.54%** for crop-specific and **11.00%** for livestock-specific, which suggests many farmers ask similar broad questions and should be able to find good existing answers faster.
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Assumptions and Caveats

- **We counted one row per question, not one row per response.** Some questions have multiple replies, so counting responses would make certain topics look larger simply because they got more replies, not because more farmers asked about them.
- **Response “specificity” is measured using keyword matching, so it is an estimate.** Keyword matching can tell whether an answer includes the

expected topic terms, but it cannot fully confirm that the answer is truly detailed, correct, or tailored the way a human reviewer could.

- **Repetition was checked only for English questions.** The dataset includes Swahili, Luganda, and Nyn, so repetition patterns could be different in those languages and are not captured by the English-only analysis.
 - **Some topics naturally overlap.** A single question can reasonably fit more than one topic, such as “maize pests” or “soil for tomatoes,” so some counts may shift slightly depending on how overlapping cases are labeled.
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Final Alignment Statement

This analysis fully addresses the Challenge 4 goal of understanding the balance between crop-specific questions and general agricultural concerns. All guiding questions related to proportions, repetition, cross-cutting topics, seasonal trends, geographic variation, and response behavior are explicitly answered using quantitative evidence and aligned with the exploratory objectives of the DataKind DataKit.